

The Runway

Every major institution across the sector is chasing the same thing: the ability to move fast on AI without losing control of risk. Some will seize this moment. Some will not — not for lack of ambition, but because they'll build on a foundation that isn't ready yet. This brief names what that foundation requires — and what it takes to build it.

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01 • WHAT WE ARE SEEING ACROSS THE MARKET

We are writing the textbook pages in real time

Across the sector, technology and risk leadership at the major institutions are navigating the same shift — different geographies, different verticals, but converging on the same inflection point.

Every major institution is navigating the same transition simultaneously: AI ambitions are accelerating, business pressure is intensifying, and the infrastructure required to govern it safely does not yet exist.

This is not one client's problem. **It is a structural shift across the entire sector — and it is happening now.**

02 • THE SHIFT THAT CHANGES EVERYTHING

From deterministic to probabilistic

For two decades, enterprise technology was built on a foundational assumption: software is deterministic. If you write the right code, the system does the right thing. Every time. Auditably. That assumption is the bedrock of how banks built their controls frameworks, compliance posture, and risk architecture.

AI breaks that assumption.

An AI agent doesn't follow a script — it reasons. A traditional application exposed to unexpected input returns an error or follows an exception path. An AI agent exposed to unexpected input makes a judgment call. That is not a marginal difference. It is the entire basis of why the existing controls infrastructure at every major Canadian bank is insufficient for where they are headed.

We are witnessing a generational shift — from **Infrastructure-as-Code to Intelligence-as-a-Service**. The building blocks are still cloud-native. But the orchestration layer is categorically new. Traditional cloud tools were engineered for deterministic software: if X, then Y. AI is probabilistic: if X, then maybe Y — with a meaningful chance of Z.

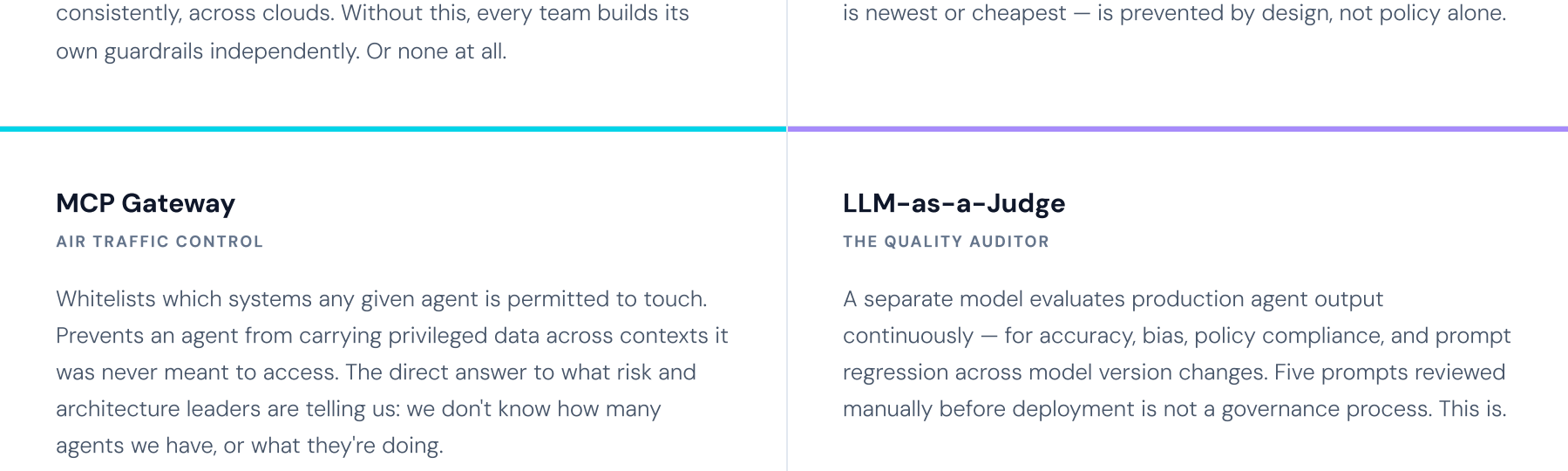
03 • THE AI LANDING ZONE

The missing operating system layer

To ground this with a frame most technology leaders will recognize: when customers began their journeys to cloud, they first built the Cloud Landing Zones — the subdivision, the land, the roads, the water, the power. VPCs, networking, storage, compute. That foundation exists across this market today.

What we are now being asked to build is **the operating system that runs on top of it** — the layer that governs how intelligence moves through the organization. How models are approved. How agents are orchestrated. How prompts are inspected. How outputs are validated. How privilege is enforced between systems.

We call this the **AI Landing Zone**. It is The Runway that enterprise AI needs before it can take flight.



It is not a single product. It is a family of capabilities — each addressing a control gap that traditional cloud architecture was never designed to fill. At this moment in time, four capabilities define the gap — and while this architecture will evolve, these are the components that matter now:

AI Gateway THE TOLL BOOTH Every prompt inspected before reaching a model. Every response validated before a user sees it. Security, cost control, and compliance guardrails enforced at a single chokepoint — consistently, across clouds. Without this, every team builds its own guardrails independently. Or none at all.	Model Registry THE LIBRARY Before any model reaches a developer, it has passed legal, ethics, and security review. Approved versions are catalogued. Ungoverned model proliferation — teams pulling whatever model is newest or cheapest — is prevented by design, not policy alone.
MCP Gateway AIR TRAFFIC CONTROL Whitelists which systems any given agent is permitted to touch. Prevents an agent from carrying privileged data across contexts it was never meant to access. The direct answer to what risk and architecture leaders are telling us: we don't know how many agents we have, or what they're doing.	LLM-as-a-Judge THE QUALITY AUDITOR A separate model evaluates production agent output continuously — for accuracy, bias, policy compliance, and prompt regression across model version changes. Five prompts reviewed manually before deployment is not a governance process. This is.

04 • THE ECOSYSTEM POWERING EACH LAYER

Each component of the AI Landing Zone draws on a maturing ecosystem of platforms and services — and the major players in this space have invested heavily to be ready for this moment. **Hyperscalers have built native AI gateway, model serving, and observability capabilities directly into their cloud platforms. Enterprise platforms are extending their data and workflow fabrics to natively support agent workloads. Specialist vendors are filling the gaps the large platforms haven't yet closed.**

HYPERSCALERS — MANAGED INFRASTRUCTURE & MODEL SERVING
AWS Azure GCP

ENTERPRISE PLATFORMS — DATA FABRIC & WORKFLOW ORCHESTRATION
Databricks ServiceNow Snowflake

The vendors themselves are actively converging toward the AI Landing Zone model — which validates the architecture thesis and accelerates the market. The institution that establishes its governance layer now will be well-positioned to adopt each platform's native capabilities as they mature, rather than retrofitting governance onto an estate that grew without it. The ecosystem is building the components. The enterprise needs to own the architecture that ties them together.

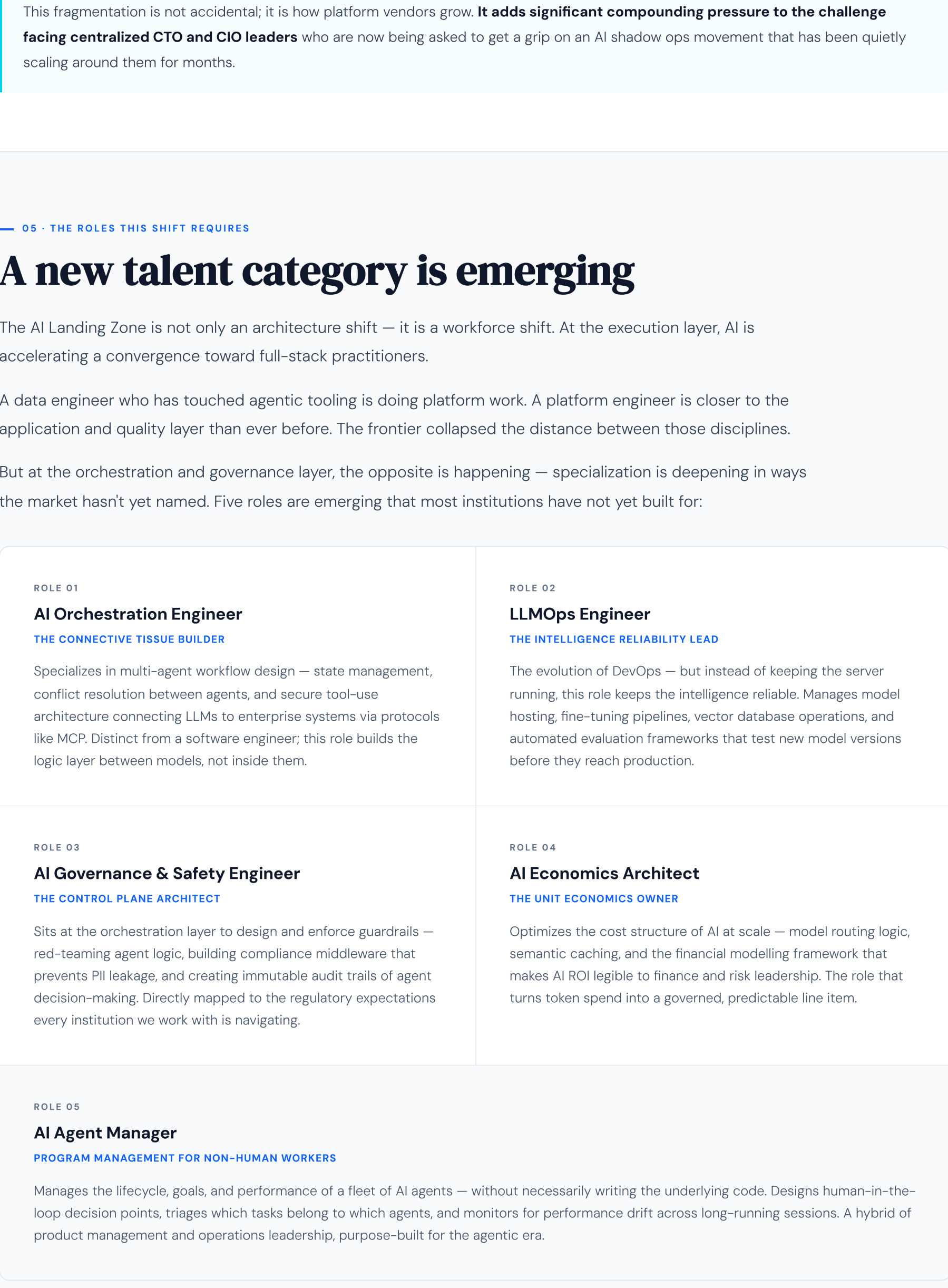
05 • MODEL ROUTING INTELLIGENCE

The AI Gateway is your cost control lever — not just your security chokepoint

Most conversations about the AI Gateway focus on security and compliance. That's necessary but insufficient. The Gateway is also the **routing brain** of the AI Landing Zone — the layer that determines which model handles each request, based on its complexity, data sensitivity, cost tier, and latency requirements.

Not every request needs a frontier model. Intelligent routing directs each task to the right model for the job — without degrading output quality where it matters. Without this layer, every request defaults to the most expensive option. The economics of AI at scale become ungovernable.

ESTIMATED COST REDUCTIONS 30–70% reduction in frontier model query costs through intelligent routing — without degrading output quality where it matters	THE DEFAULT FAILURE MODE 100% frontier, 100% of the time every AI deployment without a routing layer routes every request to the most expensive model by default	ON THE HORIZON Hybrid inference architecture local on-device execution for low-sensitivity tokens; frontier cloud only for highest-stakes decisions
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06 • THE HISTORICAL PARALLEL

Shadow AI is the new Shadow IT

In 2012, organizations were running Shadow IT — engineering teams spinning up cloud instances on personal credit cards, outside security oversight, outside procurement, outside risk. The industry solved this by building Cloud Landing Zones: centralized network architecture, IAM roles, shared services, governed access.

We are in the Shadow AI moment right now. Business units across the sector are deploying LLMs and agentic workflows without a central security handshake. Some teams have paused entirely, waiting for governance that hasn't been written yet. Others are moving anyway, accumulating risk that isn't yet visible on anyone's dashboard.

THE COMPOUNDING RISK The institution that establishes AI governance architecture now will compound that advantage for years — staffing, architecture, regulatory posture, and expansion all building on a governed foundation.	The one that waits will inherit a fragmented, ungoverned AI estate — the same way leggers inherited a decade of technical debt from unmanaged Shadow IT. The difference is that AI moves faster and the regulatory exposure is real.
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There is a compounding dynamic worth naming. Many ecosystem partners have already been running their own shadow playbooks — quietly expanding AI tooling across individual lines of business and operating units, well ahead of any enterprise governance conversation. This fragmentation is not accidental; it is how platform vendors grow. **It adds significant compounding pressure to the challenge facing centralized CTO and CIO leaders** who are now being asked to get a grip on an AI shadow ops movement that has been quietly scaling around them for months.

07 • THE ROLES THIS SHIFT REQUIRES

A new talent category is emerging

The AI Landing Zone is not only an architecture shift — it is a workforce shift. At the execution layer, AI is accelerating a convergence toward full-stack practitioners.

A data engineer who has touched agentic tooling is doing platform work. A platform engineer is closer to the application and quality layer than ever before. The frontier collapsing the distance between those disciplines.

But at the orchestration and governance layer, the opposite is happening — specialization is deepening in ways the market hasn't yet named. Five roles are emerging that most institutions have not yet built for:

ROLE 01 AI Orchestration Engineer THE CONNECTIVE TISSUE BUILDER Specializes in multi-agent workflow design — state management, conflict resolution between agents, and secure tool-use architecture connecting LLMs to enterprise systems via protocols like MCP. Distinct from a software engineer; this role builds the logic layer between models, not inside them.	ROLE 02 LLMOps Engineer THE INTELLIGENCE RELIABILITY LEAD The evolution of DevOps — but instead of keeping the server running, this role keeps the intelligence reliable. Manages model hosting, fine-tuning pipelines, vector database operations, and automated evaluation frameworks that test new model versions before they reach production.
ROLE 03 AI Governance & Safety Engineer THE CONTROL PLANE ARCHITECT Sits at the orchestration layer to design and enforce guardrails — red-teaming agent logic, building compliance middleware that prevents PII leakage, and creating immutable audit trails of agent decision-making. Directly mapped to the regulatory expectations every institution we work with is navigating.	ROLE 04 AI Economics Architect THE UNIT ECONOMICS OWNER Optimizes the cost structure of AI at scale — model routing logic, semantic caching, and the financial modelling framework that makes AI ROI legible to finance and risk leadership. The role that turns token spend into a governed, predictable line item.
ROLE 05 AI Agent Manager PROGRAM MANAGEMENT FOR NON-HUMAN WORKERS Manages the lifecycle, goals, and performance of a fleet of AI agents — without necessarily writing the underlying code. Designs human-in-the-loop decision points, triages which tasks belong to which agents, and monitors for performance drift across long-running sessions. A hybrid of product management and operations leadership, purpose-built for the agentic era.	

These roles don't yet exist in most org charts. The institutions that define and build — or contract for — this talent now will govern the agentic era. The ones that don't will manage it reactively.

Are your current role definitions, hiring pipelines, and workforce plans built for this? If not, that is a governance gap as significant as any architectural one.

08 • THE CAPABILITY FRONTIER

From execution lanes to judgment layers

The AI-native era is creating an entirely new category of institutional capability — one that sits at the intersection of enterprise architecture, regulatory fluency, and organizational design. The firms and practitioners who move early to build capability at this layer will define what great looks like for the next decade.

At the same time, the shape of delivery is evolving. Some disciplines are concentrating at the top as AI tooling absorbs routine execution work. New disciplines are forming at the foundation — **architectural, governance-oriented, and judgment-intensive** — where the competitive advantage of the next decade will be won or lost.

The opportunity is not in doing the same work faster. It's in building the capabilities that the next era of enterprise technology actually requires.

The diagram below maps where the sector is heading — and where institutional capability must be built, contracted, or accelerated now.

